Hoàng Kim Vĩnh

Embedded Firmware Engineer — Full-time Embedded / IoT roles, ideally in Japan.

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PROFILE

I write production firmware for commercial DC fast-charging stations — CAN bus, MISRA C compliance, non-blocking state machines on STM32H7 — and build complete IoT products from PCB to mobile app.

CORE CAPABILITY

Production firmware
STM32H7, CAN 2.0B, MISRA C:2012, and non-blocking state machines with a four-layer architecture.

Connected products
ESP32 and ESP-IDF firmware, MQTT, Wi-Fi provisioning, and cloud dashboards wired to real hardware.

EV charging systems
Power-sharing algorithms, OCPP charge-point management, GB/T 27930 vehicle comms, and DLT-645 metering.

Models on the MCU
YOLOv11n vision and on-device classifiers running at the edge instead of in the cloud.

EXPERIENCE

Firmware engineer, industrial EV charging — 2026

APES Lab · Hanoi University of Science and Technology

Contribute to a commercial DC fast-charging station: SPI-to-CAN driver stack, CAN frame codec, distributed power-sharing algorithms, and a fully non-blocking station state machine under MISRA C:2012.

Full-stack IoT and AIoT projects — 2025 - 2026

University capstone and enterprise internship

Delivered end-to-end IoT products — EasyEDA PCB design, ESP-IDF firmware with Wi-Fi provisioning, MQTT backends, and Flutter or Next.js applications — including the SmartHome capstone graded 10/10.

Embedded firmware intern, SmartGarden — 2024 - 2025

Yootek IoT internship

Wrote ESP32/ESP-IDF firmware for an enterprise IoT platform with a NestJS, PostgreSQL, Prisma, MQTT and WebSocket backend, including JWT auth and role-based access control.

SELECTED PROJECTS

embedded-iot-toolkit — 2026 · Firmware & Embedded

A dependency-free C99 library of the firmware building blocks I keep rewriting: a non-blocking timer bank, an ISR-safe framed serial protocol with checksums and resynchronisation, and a CAN 2.0B 29-bit codec with a bit-timing solver.

This is the public evidence of the engineering discipline the commercial firmware has to keep under NDA — a reviewer can clone it and run the tests.

· C99 · CAN 2.0B · MISRA C · cppcheck · Unity

github.com/CatKod/embedded-iot-toolkit

SmartHome — 2026 · STM32H7 + ESP32-S3

A two-tier architecture where the real-time MCU never depends on the cloud for safety. The ESP32-S3 handles Wi-Fi and MQTT; the STM32H7 owns the hardware, with TouchGFX UI, RFID entry, and an automatic window actuator.

MQTT and the dashboard can never drive hardware directly. Every command is validated by the STM32H7 before an actuator moves.

· STM32H7 · ESP32-S3 · ESP-IDF · TouchGFX · FreeRTOS · MQTT · RFID RC522

github.com/CatKod/SmartHome

Smart Power Outlet — 2026 · PCB → Firmware → App

A complete IoT product rather than a board: the PCB was designed in EasyEDA Pro, the firmware is ESP-IDF with Wi-Fi provisioning so no credential is ever hardcoded, and a cross-platform Flutter app drives it.

Graduation capstone delivered solo, covering every layer from schematic to mobile application.

· ESP32 · ESP8266 · ESP-IDF · PlatformIO · Wi-Fi provisioning · EasyEDA Pro · Flutter

github.com/CatKod/Smart_Power_Outlet

Sun-Car Damage Assessment

2026 · Edge AI for Inspection

YOLOv11n for object detection, instance segmentation and severity scoring, built to apply for a SUN scholarship. An ESP32-CAM captures frames and streams the result to an STM32 over UART, so inference sits on the edge rather than in the cloud.

Edge AI pipeline where the microcontroller receives detections over UART instead of calling a cloud endpoint.

· YOLOv11n · PyTorch · Ultralytics · ESP32-CAM · STM32 · UART · Flask · Streamlit

github.com/CatKod/Sun-Car-Damage-AI-Assessment

Yootek SmartGarden

2025 · Enterprise IoT

An enterprise IoT platform: ESP32/ESP-IDF firmware for the sensing nodes, backed by NestJS, PostgreSQL, Prisma, MQTT and WebSocket services with JWT authentication and role-based access control.

Firmware and backend written against the same device contract, so the hardware never waits on a network round trip.

· ESP32 · ESP-IDF · NestJS · PostgreSQL · Prisma · MQTT · WebSocket

github.com/CatKod/Yootek-IOT-intern

Attendance_Check

2025 · Production System

A system actually running in production at a university lab: Next.js admin panel, Electron kiosk for the terminal, an Expo mobile app, and a Supabase backend with row-level access rules.

Proves the deployment path — kiosk, mobile, and admin all read and write the same secured dataset.

· Next.js · Electron · Expo · Supabase · TypeScript

github.com/CatKod/Attendance_Check

AIoT-Face_And_Order

2025 · AIoT

ESP32-CAM captures frames and posts them to an HTTP server running OpenCV face detection, with a live web stream so the ordering flow can be driven from the detected identity.

Camera to detection to ordering handled without a cloud round trip on the hot path.

· ESP32-CAM · OpenCV · Python · HTTP

github.com/CatKod/AIoT-Face_And_Order

SmartSlide_JP

2025 · Embedded Product

A smart projector control system with a presentation remote: slide navigation, device control, and status feedback driven from a microcontroller rather than from a vendor app.

Replaces a vendor-only workflow with a system the user controls directly.

· STM32 · IR/RF · Embedded UI

github.com/CatKod/SmartSlide_JP

TECHNICAL SKILLS

- **Firmware & Embedded**
  STM32H7 firmware to the register level, CAN bus timing, MISRA-compliant drivers, and non-blocking state machines.
- **Protocols & Connectivity**
  Charging protocols, serial buses, and provisioning flows that keep the device usable without a network.
- **Hardware & Power**
  PCB layout in EasyEDA Pro, safe contactor sequencing, current ramping, and bench verification with a logic analyser.
- **Embedded GUI**
  Screen and state design for small displays driven from a microcontroller without a heavy framework.
- **Edge AI & Software**
  Model training on the desktop and inference on the device, plus the backend and app layers that connect them.
- **Quality & Process**
  Static analysis, host unit tests against a mocked HAL, and fault injection that proves the behaviour under failure.

EDUCATION

Hanoi University of Science and Technology

2022 - 2026

Information Technology

Bachelor of Engineering in Information Technology, specialising in embedded systems and industrial firmware.

CERTIFICATIONS

Japanese Language Proficiency Test — N3

CERT-01

Japan Foundation — JLPT N3 certification, issued by the Japan Foundation.

https://www.jlpt.jp/

Machine Learning Specialization

CERT-02

DeepLearning.AI × Stanford University — Supervised ML, advanced learning algorithms, recommender systems and reinforcement learning, and the capstone.

https://www.coursera.org/verify/M47TFJODCUNI

Deep Learning Specialization

CERT-03

DeepLearning.AI — Neural Networks and Deep Learning.

https://www.coursera.org/verify/XEGM5HWY79T5

Information Technology — HUST

CERT-04

Hanoi University of Science and Technology — Bachelor of Engineering in Information Technology, with embedded systems and industrial firmware focus.

https://hust.edu.vn/